Mapping Timbre Space: Can We Make More Beautiful Instruments?

Introduction

Why do people prefer certain timbres? Timbre is the quality of a sound determined by its harmonic spectrum: essentially what makes a violin sound like a violin, and a Stradivarius sound more beautiful than any old violin. It is also what allows humans to differentiate unison pitches produced by unique sources. Timbre is a critical part of composition and perception of sound, yet it remains grossly understudied and often ignored by music theorists. A possible reason for this lack of research is the physical nature of instruments; for much of history, only specific timbres could be produced by the materials on hand. However, our modern tools place within the hands of researchers and musicians timbre manipulation and control through the creation of synthetic audio with any harmonic spectrum, opening up a Wild West for timbral study.

We hypothesize that by recording people's timbre preferences, we can model the geometry of timbre space. This is an extremely exciting prospect because of its broad implications in both quantitative psychology and music. In timbre space, there may be a direction that represents beauty (and other qualities: roughness, brightness, etc.) and, if there is a direction for beauty, there may be an orthogonal direction to beauty. Understanding the geometry of timbre space would be groundbreaking for music in that more beautiful music could be produced, and in psychology it could grant critical insights into how auditory objects are perceived. For these reasons, timbre is worth investigating.

Research Background

Despite its pervasiveness in music, timbre remains relatively understudied, while adjacent auditory concepts such as consonance have been significantly more analyzed and codified (Marjieh et al., 2024). However, recent research suggests that there exist under-explored predictors for preference of musical features: the foremost of these being spectral envelope, a technical synonym for a snapshot of timbre (Eerola & Lahdelma, 2021). Timbre is a crucial part of music perception, and the primary reason it has not been studied more is that researchers were previously unable to isolate its effects. We aim to study the geometry of high-dimensional timbre space by recording timbre preferences and timbre's interaction with other features of auditory preference, which are recognized as roughness, harmonicity, and familiarity (Eerola & Lahdelma, 2021).

The oldest recognized predictor of preference for auditory objects is roughness. Early studies on roughness investigated preferences of intervals between two pure tones (tones without harmonics). It was originally hypothesised that there exists a critical bandwidth of intervals around a tone for which its interaction with another tone is perceived as rough (Plomp & Levelt, 1965). From this, further investigation showed that within this critical bandwidth "beats" could

be heard, caused by the rapid oscillation of interaction between the two tones (Plomp, 1967). This beating has become the strongest canonical predictor of preference of musical features.

The measurement of perception of beats was later codified into the more objective measure of tonal fusion. When beats are not present, two tones are more likely to be perceived as one (DeWitt & Crowder, 1987). This is a more measurable statistic than perceived roughness and is often used as a proxy for roughness. Recently, it has been shown that timbre has an interaction with beating; that under specific timbres the range for which beats are heard varies, and intervals which are canonically consonant (or smooth) can be perceived as rough (Madsen & Oxenham, 2024). This, in itself, is a significant reason to further investigate timbre.

The second feature for auditory preference, harmonicity, is intrinsically tied to roughness. Harmonicity is the property of sound having harmonic frequencies that sound simultaneously with the fundamental frequency. Tones with harmonicity are called "complex tones." By definition, audio with timbre has harmonicity. Perceptions of roughness for complex tones have been recorded as being approximately the sum of roughness created by the interval between each pair of harmonics (Plomp & Levelt, 1965). In addition to affecting perceptions of roughness, harmonicity can convey enough information about the fundamental pitch of a complex tone that melodies and harmonies of complex tones can be recognized without their fundamental pitch (Houtsma & Goldstein, 1972). Both of these discoveries indicate that harmonicity – by definition related to timbre – has a significant impact on auditory perception.

Finally, familiarity. It has recently been shown that exposure to Western music impacts preference for harmonicity. One study showed that individuals with increased musical training in the US are much more favorable to harmonicity (McDermott et al., 2010), and another recent study showed that, although tonal fusion is constant across cultures, preference for harmonicity and canonical consonance are purely a function of exposure to Western music (McPherson et al., 2020). The impact of familiarity presents an interesting avenue for study of its interaction with timbre, especially with the newness of these results.

Although timbre has been recently recognized as an important part of music perception, there are limited studies on it and its interaction with other features of music. Because of the potential value of mapping timbre space in discovering more beautiful timbres – equivalent to more beautiful instruments, it is worthwhile to investigate timbre.

Methodologies

Participants

For this study, we will recruit a large and diverse sample of participants. We will need a large sample size for the study to be adequately powered (Wallisch, 2015). We will recruit participants both from the NYU subject pool (SONA) for course credit and from the general community for monetary compensation. Participants will complete a pre-screening survey before starting the study to record demographic information and musical background.

Materials and Measures

Each participant will listen to brief (5-second) clips of 100 melodies, selected at random from a larger pool of 1,000 candidate melodies that cover the entire range of music across time, genres, and popularity. It has previously been shown that 5-second clips of music are representative of entire works (Philibotte et al., 2023). Each melody will have a corruption function applied to its audio that changes its timbre. The participant will be presented with a one-dimensional continuous slider that applies a scalable transformation (e.g., scrunching, stretching, raising, etc.) to the harmonics of the corrupted melody, with a specific choice returning the audio to its uncorrupted form. The clip will repeat continuously, and the participant will be asked to adjust the slider to maximize a specific audio quality (e.g., beauty, roughness, brightness, etc.). This will continue for 30 seconds or until the participant is satisfied, whichever comes first. They will then rate their enjoyment of the clip on a 5-point Likert scale ranging from "Love it" to "Hate it," and rate their familiarity with the clip on a 3-point scale from "never heard before" to "heard often." Each clip will be interspersed with a "palate cleanser" trial, either an assessment of current mood (every 20 clips) or an item from a personality instrument. This will help to minimize spillover effects between the clips. Importantly, it will also allow us to assess their personality, mood, and mental health. Specifically, we will use items from the TIPI (Gosling et al., 2003), PHQ9 (Williams, 2014), and Dirty Dozen (Jonason & Webster, 2010).

Procedure

Our participants arrive at the testing location and are provided with a form to obtain written informed consent from research assistants. We will also confirm that they have completed our pre-screening survey. Once this is confirmed, they will be walked to the testing room, where they are seated at a computer and given a set of headphones. Research assistants will demonstrate how the study works by going through a short training module and also set the sound to a comfortable volume. Research assistants then leave the participants to complete the study on their own, but are available outside the room should there be any questions. After completing the study, participants are fully debriefed as to the purpose of the study.

Analysis

We will use a neural network to find which harmonic features and harmonic feature interactions with personal features (i.e., musical familiarity, psychological profile) are most predictive of large errors from the uncorrupted audio. We will measure the predictive loss of this neural network using test and validation sets of songs the network has not been trained on – measuring error in its predicted optimal scalar transformation. Because weights of neural networks are uninterpretable, we cannot directly analyze the predictive power of each feature; so we will use gradient descent on the output of the network to find timbres that optimize or minimize certain qualities. We will analyze these timbres and use them to make generalizations about timbre space. We will also use a non-linear dimensionality reduction, such as t-SNE, to understand how certain qualities of music are similar in timbre space.

Timeline

Our research project is aimed to start in the first week of November and run through the last week of April 2027. We anticipate about 10 hours per week for each member to be dedicated to the work, with a total of 30 hours per week. This effort breaks down as follows: Each week, we will record data from 10 participants, taking 20 hours; with no-shows, about 25 hours. The rest of the time, we will spend on data analysis and modeling. In March, the project will transition into full-time writing and presentation mode, as we plan to present this work at a scientific conference and also prepare it for publication.

Motivations

Craig is a Data Science major, minoring in Music, Math, and Economics. Through this, he provides strong analytical skills and music domain knowledge. He is interested in pursuing a career in research, so this project would be helpful in advancing that goal. Jack Williams is at Gallatin, concentrating in the neuroscience and psychology of art and music, and film music composition. Through this, he provides domain knowledge on music and music technology, and perceptual neuroscience and psychology. This project is especially relevant to his goal of being a music cognition researcher. Katherine is a pianist and researcher specializing in musical temperament and timbral perception. Her interdisciplinary background in performance, tuning theory, and music cognition brings valuable musical and perceptual expertise to the project, which aligns closely with her research on how subtle variations in harmonic spectra shape consonance and the perceived character of sound.

Relationship with PI

Craig met Professor Maloney through Professor Pascal Wallish, for whom Craig works as a research assistant, in the Spring 2026 semester. In this meeting, Craig was introduced to this research idea and the other members of the team.

Katherine's collaboration with Professor Maloney began during her doctoral research on temperament. Their current psychoacoustic investigations into how harmonic spectra and timbre influence consonance perception directly informed the development of this project.

Jack Williams' work with Professor Maloney and Dr. Lin began during his first semester at NYU in 2024, where he was invited to join their group and quickly started developing experimental methodologies for novel investigations into music. Jack is a Gallatin student who is

a multi-instrumentalist, composer, and producer with music technology and audio engineering domain knowledge. In 2025, Jack led a pilot study related to the current work, creating an interface and synthesizer for studying timbre and tuning systems, and furthering research related to Dr. Lin's thesis and her work with Maloney.

We developed this project closely with Professor Maloney, and under his guidance, we currently meet weekly to refine experimental design. During participant testing, we plan to meet twice weekly to review progress.

Costs

As we aim for high statistical power (Wallisch, 2015), we will need to attain a large sample size. Whereas we will recruit some of our participants from the NYU SONA pool for class credit, it is also important to us to recruit a diverse pool of participants who are not NYU students. These participants will need to be paid at the minimum wage rate of $15 per hour in New York. Therefore, we request a budget of $1250 to recruit 40 participants, as each participant will have to be paid for 2 hours, allowing some money to pay for overtime in cases where some participants will need more time, as our piloting suggests they will.
The payment of these participant fees is our only anticipated expense.

Duties

Every week, each member will assist in data recording for 6 hours, weekly lab meetings for 2 hours, and time allocated for data analysis of 2 hours each, summing up to a total of 10 hours. Data recording will be done in shifts of 2 hours by running participants who are scheduled at the start of each week. Given Craig's strong background in data science and math, he will lead the quantitative analysis of the recorded data. Katherine has a rich musical background and therefore will spearhead the implications of results for music. Finally, Jack has a breadth of knowledge in psychology and neural science, and therefore will interpret what the data says about auditory perception. Overall, our team is very well fit for this project, given we have an expert in each of its dimensions.

Works Cited

Eerola, T., & Lahdelma, I. (2021). The anatomy of consonance/dissonance: Evaluating acoustic and cultural predictors across multiple datasets with chords. *Music & Science, 4*. https://doi.org/10.1177/20592043211030471

DeWitt, L. A., & Crowder, R. G. (1987). Tonal fusion of consonant musical intervals: The Oomph in Stumpf. *Perception & Psychophysics, 41*(1), 73–84. https://doi.org/10.3758/BF03208216

Gosling, S. D., Rentfrow, P. J., & Swann, W. B., Jr. (2003). A Very Brief Measure of the Big Five Personality Domains. Journal of Research in Personality, 37(6), 504-528. https://doi.org/10.1016/S0092-6566(03)00046-1

Houtsma, A. J. M., & Goldstein, J. L. (1972). The central origin of the pitch of complex tones: Evidence from musical interval recognition. *Journal of the Acoustical Society of America, 51*(2B), 520–529. https://doi.org/10.1121/1.1912873

Jonason, P. K., & Webster, G. D. (2010). The dirty dozen: a concise measure of the dark triad. *Psychological Assessment*, *22*(2), 420-432. https://doi.org/10.1037/a0019265

Madsen, S. M. K., & Oxenham, A. J. (2024). Mistuning perception in music is asymmetric and relies on both beats and inharmonicity. *Communications Psychology, 2*, Article 91. https://doi.org/10.1038/s44271-024-00141-1

Marjieh, R., Harrison, P. M. C., Lee, H., Deligiannaki, F., & Jacoby, N. (2024). Timbral effects on consonance disentangle psychoacoustic mechanisms and suggest perceptual origins for musical scales. *Nature Communications, 15*, Article 1482. https://doi.org/10.1038/s41467-024-45812-z

McDermott, J. H., Lehr, A. J., & Oxenham, A. J. (2010). Individual differences reveal the basis of consonance. *Current Biology, 20*(11), 1035–1041. https://doi.org/10.1016/j.cub.2010.04.019

McPherson, M. J., Dolan, S. E., Durango, A., Ossandon, T., Valdés, J., Undurraga, E. A., Jacoby, N., Godoy, R. A., & McDermott, J. H. (2020). Perceptual fusion of musical notes by native Amazonians suggests universal representations of musical intervals. *Nature Communications, 11*, Article 2786. https://doi.org/10.1038/s41467-020-16448-6

Philibotte, S. J., Spivack, S., Spilka, N. H., Passman, I., & Wallisch, P. (2023). The whole is not different from its parts: Music excerpts are representative of songs. *Music Perception: An Interdisciplinary Journal*, *40*(3), 220-236. https://doi.org/10.1525/mp.2023.40.3.220

Plomp, R., & Levelt, W. J. M. (1965). Tonal consonance and critical bandwidth. *Journal of the Acoustical Society of America, 38*(4), 548–560. https://doi.org/10.1121/1.1909741

Plomp, R. (1967). Beats of mistuned consonances. *Journal of the Acoustical Society of America, 42*(2), 462–474. https://doi.org/10.1121/1.1910602

Wallisch, P. (2015). Brighter than the sun: Powerscape visualizations illustrate power needs in neuroscience and psychology. arXiv preprint arXiv:1512.09368. https://doi.org/10.48550/arXiv.1512.09368

Williams, N. (2014). PHQ-9. *Occupational Medicine*, *64*(2), 139–140. https://doi.org/10.1093/occmed/kqt154